Business Case 2:

Predictive Analysis of Financial Performance

For the automotive industry

Report Team 1

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Work allocation

For the case of "Predictive Analysis of Financial Performance for an automotive industry", Team 1 has decided to divide the work as follows:

- **Data collection** and **processing using Machine Learning** algorithms by Alessia Ferrero and Antonio Castriotta
- **KPIs** and **Financial Ratios** developed by Pham Minh Duc and Alessandro Falcetta
- **Mockup of the website** was created by Francesco Capria
- **Collection of results** and **creation of the report** were handled by Giulia Mela and Ilenia Fortuna
- **Presentation's slides** and **timeline** were created by Giulia Mela, Ilenia Fortuna and Hritik Padwal

Introduction

This report presents an extensive predictive analysis of Ferrari's financial and operational performance, using historical data collected from 2014 to 2023, sourced from AIDA and Ferrari's official annual reports. The analysis applies machine learning models and statistical techniques to estimate key performance indicators (KPIs) and forecast 2024 metrics.

Our goal is to provide actionable insights that will aid in predicting cash flows, budget needs, production planning, and procurements more accurately. This predictive analysis also considers the impact of economic conditions on Ferrari's financial performance, similar to how automotive manufacturers adjust their operations based on market trends and sales data. This analysis has been conducted by **Pisa Consulting**, a leading consultancy firm specializing in data-driven business solutions and financial forecasting.

1. Goals

The goals that **Pisa Consulting** has set to achieve in this business case are as follows:

1. Predict Ferrari's financial metrics, including revenues, costs, and inventories, for 2024.
2. Estimate KPIs such as Revenue Growth Rate, Gross Profit Margin, Operating Profit Margin, and R&D Rate for 2024.
3. Forecast critical financial ratios such as Return on Sales (ROS), Return on Assets (ROA), Return on Equity (ROE), and Return on Investment (ROI) for 2024.
4. Provide insights into historical trends and their implications for Ferrari's future performance.

2. Methodology

The analysis employed the following methods:

1. **Machine Learning:** Weighted Linear Regression to forecast metrics for 2024.
2. **Statistical Analysis:** Calculation of KPIs and prediction of financial ratios (ROS, ROA, ROE, ROI) for 2024.
3. **Mock-up:** Visualization of the forecasting software.

3. Results and Analysis

3.1 Financial Metric Predictions (2024)

Metric	Prediction for 2024 (in million)
Net Revenues	6328
Cost of Sales	3250
Selling, General, and Administrative Costs	573
R&D Costs	512
Depreciation and Amortization	343
Cash Flow from Operating Activities	629
Cash Flow from Investing Activities	-386
Cash Flow from Financing Activities	292
Work-in-Progress Inventories	263
Finished Goods Inventories	262
Raw Materials Inventories	223
Car Sales (in units)	17075

The data presented in this table provides a general overview of the financial and operational performance forecasted for 2024. From this, we can draw several observations:

1. **Sustained Growth with Cost Focus**
   - The projected net revenues (6,328 million) indicate an expansion of business activities, likely supported by increased sales or an effective commercial strategy. However, the relationship with the cost of sales (3,250 million) suggests that the company must maintain careful management of operating margins to ensure profitability.

2. **Efficient Control of Operating Expenses**
    - General and administrative expenses (573 million) and R&D costs (512 million) are well-managed, showing moderate growth relative to the overall scale of the business. This highlights a good balance between cost containment and strategic investment.
3. **Stable Cash Flow**
    - The positive operating cash flow (629 million) demonstrates the company's solid ability to generate liquidity from its core activities, despite significant investments (cash flow from investing activities: -386 million). The positive financing cash flow (292 million) may indicate capital raising or strategic reinvestment.
4. **Inventory Management**
    - The forecasts for inventories (raw materials, finished goods, and work-in-progress) suggest efficient production planning and logistical operations. This is crucial to meeting demand without overburdening costs.
5. **Expanding Sector**
    - The forecasted car sales (17075 units) reflect consistent and stable growth, aligned with revenue trends. This could be an indicator of sustained demand, market expansion, or an updated product portfolio.

3.1.1 Graphical Overview

Historical Trend of Net Revenues with 2024 Forecast

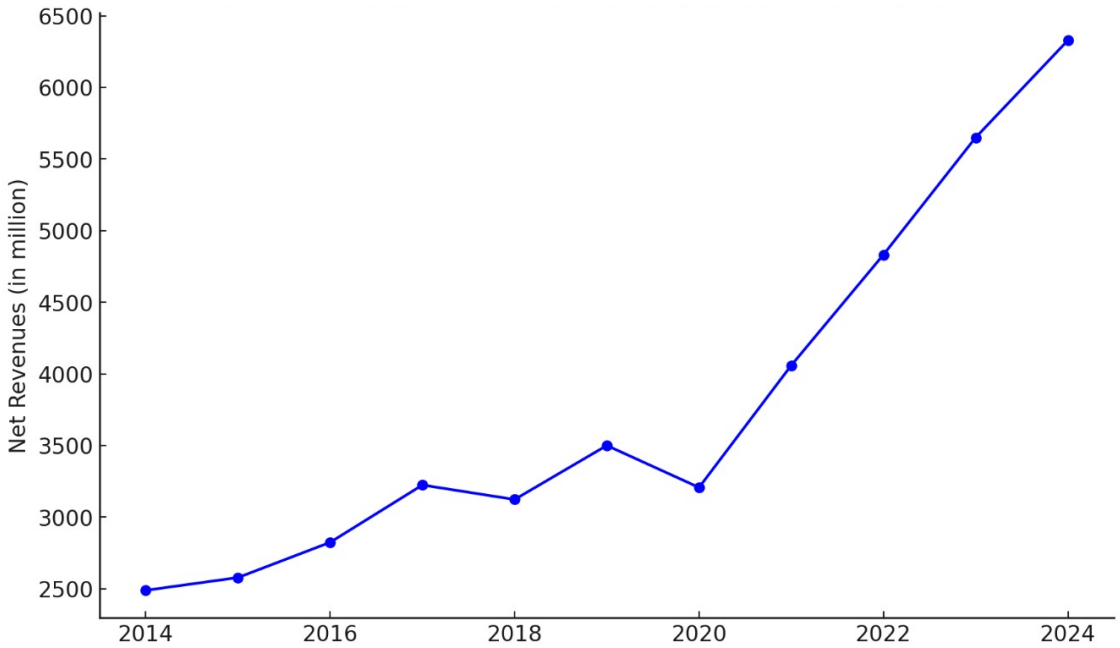


Fig 4.1 Historical Trend and Forecast of Net Revenues (2014–2024)

The chart displays the historical trend of net revenues from 2014 to 2024. A steady growth is observed over the years, with a single dip in 2020, likely attributable to the effects of the COVID-19 pandemic. The decision to use a weighted linear regression, despite its inherent limitations, was driven by the fact that in the years following the COVID-19 period, the linear model proved particularly effective in capturing the trend of the data. By assigning greater importance to more recent years, the weighting

allows for a better representation of the company's recovery dynamics and adaptation to new market conditions. This approach reflects the strong growth observed in recent years and provides a more accurate projection of future performance.

Cost of Sales from 2014 to 2024

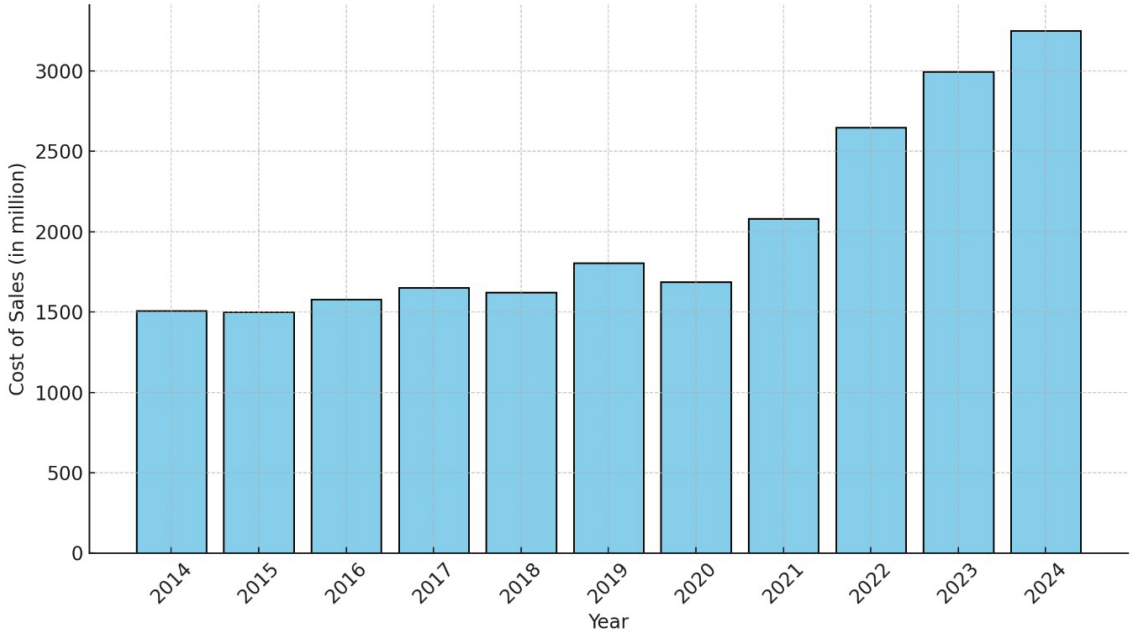


Fig 4.2 Historical Trend of Cost of Sales (2014–2024)

The bar chart illustrates the historical trend of the company's Cost of Sales from 2014 to 2024. The costs remain relatively stable from 2014 to 2020, showing only minor variations. A notable increase begins in 2021, aligning with the company's post-pandemic recovery and expansion. The estimates for 2024 indicate a continued upward trajectory, reflecting higher production and operational costs associated with growing sales volumes. This trend underscores the company's focus on scaling activities while highlighting the importance of cost management to maintain profitability.

2024 Expense Distribution

The pie chart represents the breakdown of the company's major expenses projected for 2024, divided into three main categories: **Cost of Sales** (75.0%), **Selling, General & Administrative** (SG&A) (9.6%), and **Research & Development** (R&D) (8.6%). The predominance of the Cost of Sales highlights the centrality of operational activities within the cost structure, while the expenses for R&D and SG&A demonstrate the company's commitment to innovation and strategic management of operations.

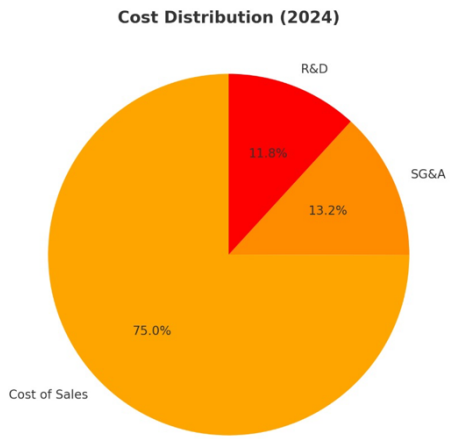


Fig. 4.1 Percentage Distribution of Expenses for 2024

3.2 Key Performance Indicators (KPIs)

In the table below there are historical and forecasted KPIs, including predictions for 2024.

Year	Revenue Growth Rate (%)	Gross Profit Margin (%)	Operating Profit Margin (%)	R&D Rate (%)
2014	18.29	45.51	15.10	19.55
2015	3.33	47.51	16.01	19.66
2016	8.79	49.15	19.90	19.74
2017	10.02	51.70	22.83	19.23
2018	0.12	52.57	24.21	18.80
2019	10.12	52.07	24.40	18.56
2020	-8.15	51.26	21.10	20.44
2021	23.45	51.29	25.15	17.99
2022	19.32	48.03	24.44	15.21
2023	17.17	49.83	27.34	14.76
Predicted 2024	6.00	48.64	31.49	8.09

- **Revenue Growth Rate (%):** The revenue growth has difference year by year, with a big jump of 23.45% in 2021, likely market recovery. However, the predicted growth of 6% in 2024 suggests possible challenges, such as increased geopolitical competition.
- **Gross Profit Margin (%):** The gross profit margin has improved, reaching 49.83% in 2023, showing strong cost management. The slight expected drop to 48.64% in 2024 suggests the need to stay focused on controlling costs.
- **Operating Profit Margin (%):** Operating profit margins have been increasing, reaching 27.34% in 2023. The forecasted rise to 31.49% in 2024 shows better efficiency and the company benefiting from improved operations.
- **R&D Rate (%):** The R&D investment has decreased from 19.55% in 2014 to 14.76% in 2023, showing more efficient spending. The expected drop to 8.09% in 2024 suggests the company might focus more on profitability than heavy investment in innovation during a tough market period.

3.3 Financial Ratios

This table presents Ferrari's historical values for ROS, ROA, ROE, and ROI from 2014 to 2023, along with the forecasted values for 2024.

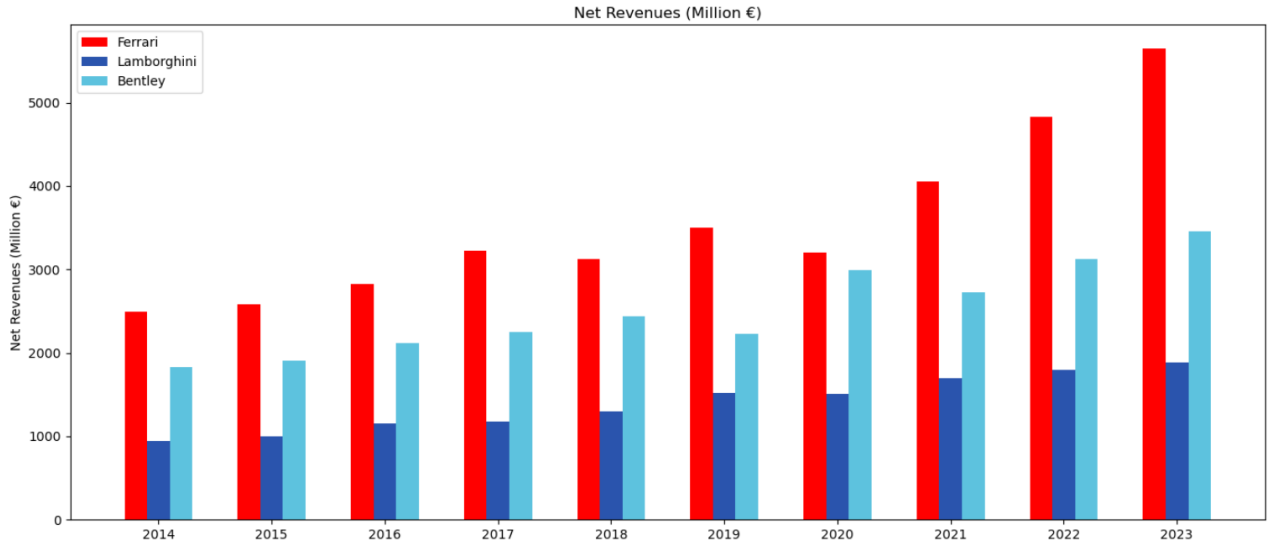
Year	ROS (%)	ROA (%)	ROE (%)	ROI (%)
2014	13.33	12.75	20.08	11.69
2015	14.91	12.65	18.16	12.82
2016	18.59	14.28	19.00	13.92
2017	22.25	18.52	22.76	15.00
2018	18.45	13.49	22.78	16.05
2019	20.94	16.13	21.66	17.07
2020	16.92	10.54	12.33	12.17
2021	20.83	14.20	17.08	14.38
2022	21.91	16.50	20.67	12.08
2023	25.99	20.16	24.86	15.38
2024 (Predicted)	26.15	23.78	27.18	14.62

- **Return on Sales (ROS)**: The ROS has shown a consistent upward trend, reaching 25.99% in 2023, reflecting strong profitability. The forecasted slight increase to 26.15% in 2024 suggests that Ferrari is maintaining robust cost controls and achieving higher profitability relative to its sales.
- **Return on Assets (ROA)**: ROA has improved steadily over the years, peaking at 20.16% in 2023. The predicted increase to 23.78% in 2024 indicates enhanced efficiency in utilizing assets to generate profits, potentially driven by optimized operations and effective asset management.
- **Return on Equity (ROE)**: ROE has shown solid growth, reaching 24.86% in 2023. The forecasted increase to 27.18% in 2024 highlights Ferrari's ability to deliver higher returns to its shareholders, possibly driven by strong revenue growth and efficient capital structure management.
- **Return on Investment (ROI)**: ROI has fluctuated over the years, achieving 15.38% in 2023. The slight forecasted decrease to 14.62% in 2024 suggests stable but slightly reduced profitability from investments, possibly due to higher capital expenditures or a more competitive market environment.

4. Global Market Analysis: Ferrari vs. Competitors (2014–2023)

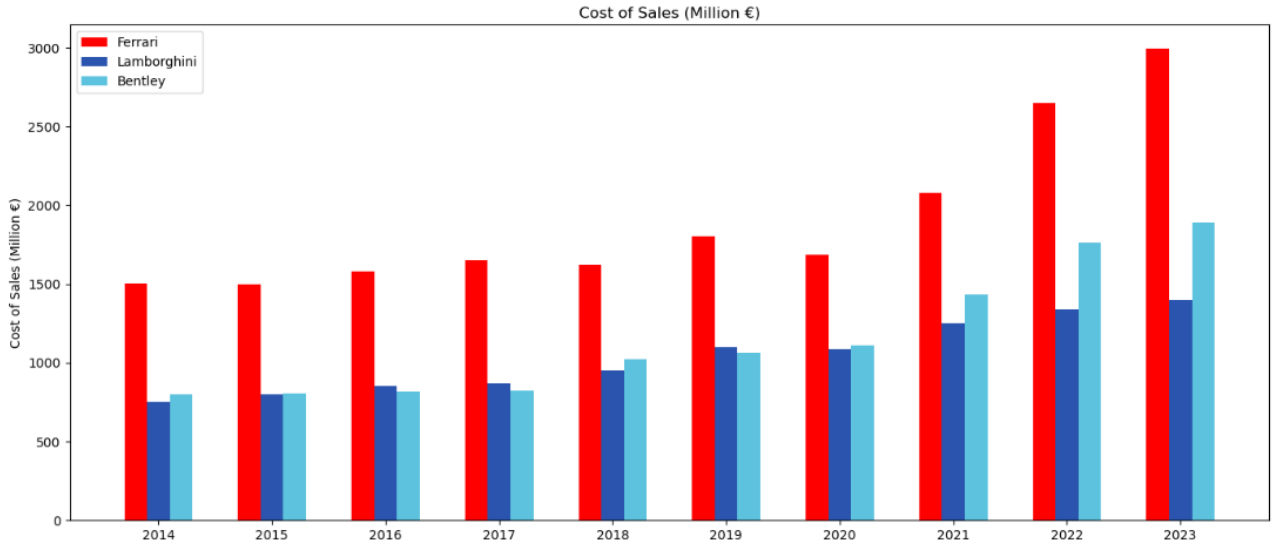
4.1 Net Revenues

- **Ferrari**:
  - Ferrari's net revenues have shown steady growth, rising from €2.489 million in 2014 to €5.650 million in 2023.
  - Ferrari maintains a strong position due to a combination of high exclusivity, brand heritage, and a diversified model lineup (including recent hybrid engines).
- **Lamborghini**:
  - Lamborghini is a strong competitor, with revenues growing from €940 million in 2014 to €1.890 million in 2023.
  - However, Lamborghini's revenues are lower than Ferrari's, positioning it in a smaller market segment. Growth has been driven by models such as the Urus, which boosted sales in non-traditional segments (SUVs).
- **Bentley**:
  - Bentley's revenues are relatively stable but modest, increasing from €1.800 million in 2014 to €3.400 million in 2023.
  - Its market focus is on elegance and comfort rather than high performance, positioning it differently from Ferrari and Lamborghini.



4.2 Cost of Sales

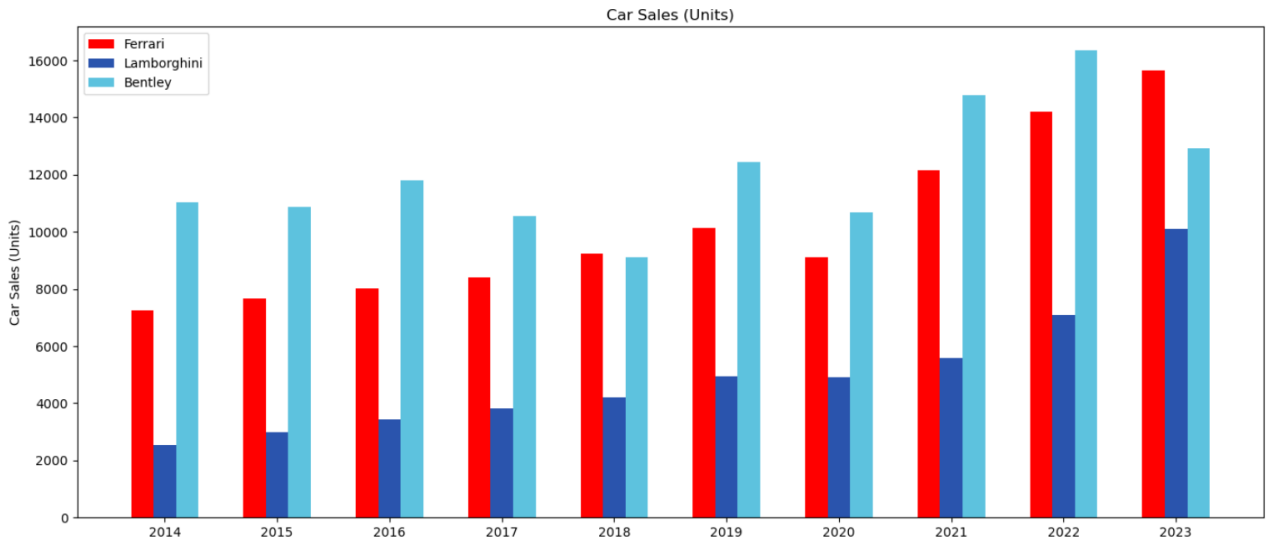
- **Ferrari**:
  - Ferrari maintains effective control over production costs, which increased from €2.762 million in 2014 to €2.995 million in 2023. This indicates relatively stable margins due to production optimization and exclusivity strategies.
- **Lamborghini**:
  - Lamborghini's production costs have grown in line with revenues, rising from €750 million in 2014 to €1.400 million in 2023. Lamborghini's strategy leans toward higher production volumes to meet demand, especially in the SUV segment.
- **Bentley**:
  - Bentley's production costs are relatively low compared to the other two brands, starting at €798 million in 2014 and reaching €1893 million in 2023. Its production is more limited, focusing on craftsmanship and uniqueness.



4.3 Car Sales

- **Ferrari**:
  - Ferrari's car sales have grown from 7,255 units in 2014 to 15,665 units in 2023, doubling in a decade.
  - This reflects increasing global demand for iconic models like the Ferrari SF90 Stradale and its expansion into emerging markets.
- **Lamborghini**:
  - Lamborghini's car sales have risen significantly, from 2,530 units in 2014 to 10,112 units in 2023. This growth has been driven by the success of the Urus SUV, which accounts for over 50% of annual sales.
- **Bentley**:

- Bentley's car sales have shown fluctuating growth, peaking at 16,360 units in 2022 and dropping to 12,940 units in 2023. Its focus on luxury comfort rather than high performance limits its direct competition with Ferrari and Lamborghini.



4.4 2024 Ferrari vs. Competitors forecast summary

For 2024, Ferrari, Lamborghini, and Bentley are expected to continue their growth in the luxury automotive market:

Ferrari: Forecasted revenues of €6,328 million with 17075 cars sold, driven by exclusive model launches and strong brand appeal.

Lamborghini: Predicted revenues of €2,117 million and 11629 cars sold, supported by the popularity of its SUVs and supercars.

Bentley: Expected revenues of €3,963 million with 13457 cars sold, benefiting from its focus on luxury craftsmanship and hybrid innovations.

Ferrari remains the leader in exclusivity and margins, while Lamborghini and Bentley capitalize on expanding product lines and emerging markets. These estimates reflect their resilience and adaptability in the luxury car segment.

Metric (Prediction 2024)	Ferrari (2024)	Lamborghini (2024)	Bentley (2024)
Net Revenues (Million)	6,328	2,117	3,371
Car Sales (Units)	17075	11629	13457
Cost of Sales (Million)	3,250	1,540	2,103

5. Final Considerations

In 2024, Ferrari will continue to maintain a strong position in the luxury car market. With forecasted **revenues** of €6.328 billion and **sales** of 17,075 units, the company will show moderate growth, though slightly slower than in previous years. The increase in some profitability indicators, such as **ROS** (from 25.99% to 26.15%) and **ROA** (from 20.16% to 23.78%), suggests improvements in operating margins and efficiency in resource utilization. The company still maintains a solid profit base. **Cost of sales** are expected to be €3.250 billion, with strict management of operating expenses reflected in **general and administrative** costs (€573 million) and **R&D** costs (€512 million). These investments in research and development are crucial for maintaining innovation and competitiveness, supporting the continuous evolution of Ferrari's exclusive models, such as new hybrid versions. The **operating cash flow**, forecasted at €629 million, highlights the company's strong ability to generate liquidity, despite investments in operating activities and a negative **cash flow from investments** (-€386 million), suggesting that Ferrari will continue to pursue growth opportunities and improve its production capabilities. Ferrari faces challenges in an evolving market, but thanks to its strategy based on exclusivity, high quality, and innovation, it will maintain its leadership position in the luxury car sector.

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